

Laetitia Sonami

Laetitia Sonami is a composer, performer, and sound artist born in France in 1957 and now based in Oakland, California. Since 1991, she has designed and built new gestural controllers for musical performance and composition. Her main instrument is the Lady's Glove, a glove modified with sensors, which allows her to use subtle movements of each finger to control sounds, mechanical devices, and lights in real time. She created the first one in 1991 for the Ars Electronica festival in Linz, Austria, and has developed four updated versions since then. With the Lady's Glove, Sonami has performed many works for voice and electronics, several of which use texts by Melody Sumner Carnahan as source material. Her compositions have been described as "performance novels," because musical form and textual narrative unfold and are transformed through her physical motions.

Sonami also makes sound installations that combine audio and kinetic elements embedded in ubiquitous objects such as light bulbs, rubber gloves, bags, and toilet plungers. Most recently, she has collaborated and toured with Sue Costabile, creating live films by combining sound, video, and kinetic elements in performance. Her work was often discussed when I was a graduate student at Mills, since she had gone through the same program in the late '70s and still lives and works nearby. We met for this interview in May 2004 at her home in Oakland.

:::

Tara Rodgers: How did you get started building your own instruments?

Laetitia Sonami: It was really when I came here, to Mills College, in 1978. I first went to the Boston Museum School of Fine Arts. I was just

seventeen and planned to study visual arts. They had two Putney VCS3 synthesizers, nice, portable, small synths. They did not sound that great, but they were not too expensive and were available. One day a friend said I should check them out, and I was like, Wow, what is that? I was hearing sounds that were very crude, but still there was this whole sense of magic, of electricity producing sounds in ways I could not fathom. I had learned the piano as a kid, but up to that point, music or sounds were not a preoccupation. So it was a surprise, discovering this whole new world and being mostly puzzled by the mystery, not understanding how it worked. I realized I didn't even understand how sounds were formed, how they blended together.

I went back to France, and I tried to go to the GRM [Groupe de Recherche Musicale] where they had a lot of electronic music and recording equipment, and they said, No, you have to go to the conservatory for two years. But I really was not coming at it from an idea that it was a continuation of musical ideas. I was interested in the technology and understanding sounds, and I didn't think I had to learn harmony to be able to do it. I can see why some people would, and expand their vocabulary with technology, but this was not my situation.

So that is when I met Eliane Radigue. A friend of mine knew the owner of a very experimental record label called Shandar, and she knew Eliane, and suggested I meet with her. After first meeting with Eliane, she allowed me to use her equipment (a wonderful ARP 2500) two, three times a week and see what would come out of it. We became very close friends, one of those friendships that change the directions of your life. She ended up letting me work there all year. Once a week she would leave—she just had a small place—and I would work, and then she would come back and we would talk and eat and listen. She knew I couldn't really do my work in France; it had quite a rigid system which required you to follow particular schools of musical thought. She was herself a student of Pierre Henry, and had encountered difficulties when it became clear that her music had nothing to do with that of her mentor. He was the great master of *musique concrète*, and her music was very subtle, with slowly evolving sonic spaces. So, to make a long story short, Eliane introduced me to Joel Chadabe. He had this incredible Moog studio at the State University at Albany, so I studied with him for one year. And then I came to Mills in '78, and ended up studying with Robert Ashley and David Behrman.

Mills had an incredible pool of self-taught builders, such as Paul

DeMarinis, John Bischoff, Frankie Mann, Rich Gold, and many others. They were building their own homemade, personal electronic systems. It was expected that you would build what you needed, or at least try! And I was shocked: build my own musical tools? It was definitely not in the European tradition. There you have engineers and artists, and there is a very clear separation. While here, definitely, the separation between engineers and artists, at least at Mills, but I think in California and America in general, is more fluid, so the feedback between art and technology is much tighter. So that was quite a surprise. They showed me how to order a kit, put it together . . . I never got very good at it, but I got to be not scared at blowing things up! I figured then that you needed to know at least enough to do it wrong, and then people will always say, How come you didn't do it that way? And show you how to improve. But if you don't try, nobody will show you. It's like a car, if you show a guy your car, and you fixed something wrong, they're always going to try to fix it for you! So I figured with electronics, it would be a good approach. Always do something, and even if it's totally wrong, there will be ten guys saying, That's not the way to do it! [Laughs]

But I like building, and using my hands, even more now that we are so immersed in digital systems embedded in black boxes. I try to get students to do it, and people really like it. It is very popular right now, hacking, building . . . It's funny, one of the classes I teach is Digital Audio Systems, and one or two times a semester we build analog circuits. You would think with all this equipment, and all the things you can do with digital audio, that it would keep their interest. But nothing compares to building a little oscillator that goes *wooo-wooo-wooo*! People are amazed, because they can see a connection to this object that produces something unexpected. It is harder to do with a computer, where everything is out of sight. With a circuit, you still have this idea of a wire that goes to this, which goes to that, and then to this . . . There is a palpable causality that you don't understand, but is left to be deciphered, like magic.

When did you build the first Lady's Glove?

The first Lady's Glove was in '91. It was a pair of rubber gloves for a performance with Paul DeMarinis for Ars Electronica. The second one came soon after, and the third one, in '93, was much more evolved. The fourth one I used from '97, for six years. And now I'm using the fifth

version, which I finished in December 2003. Bert Bongers, who was sponsored by STEIM in Amsterdam, built the last two versions.

The glove was first more of a joke, a kind of a social commentary on technology, but then it became an instrument. I've been trying to figure out at which point a controller becomes an instrument . . . I think that when you use or design a controller, and if you're just using it to push buttons or trigger things, it does not really affect the way you think of the music or how you write the software. You have your ideas and you're using a controller as an interface. Then I would not call it an instrument. I think it becomes an instrument when the software starts reflecting and adapting the limitations and possibilities of the controller, and your musical thinking and ideas become more a symbiosis between the controller, the software, and the hardware. I used to have much more of a distinction between the hardware, my sounds, the software, and the gestures—a sort of puzzle. Now I realize that my imagination is pretty much molded by the system I use. I don't think as much how will I adapt my ideas to the instrument, but I realize that the instrument has already influenced what I envision. So the Lady's Glove is more of an instrument now, good or bad.

And with it, sounds in a way are embodied through your gestures.

I think so. Recently someone asked me whether the system is imposing itself, or I'm controlling it. It's a little of everything. I sometimes dream it could be inside my body, grafted in the skin. But then, I would miss the awkwardness of an external apparatus, a mechanical system that the body's trying to adapt to, and the struggle that comes with it.

But I got a little tired of gesture controllers. I used to think: laptop performers, they never think about movement, they're so boring. But now I actually quite envy them, in a way. That ability to ignore an audience is very tempting to me. I'm always very aware of the audience, and of performance. On my dark days I feel like I'm just a glamorous waitress! [Laughs] This idea of service . . . There's this traditional, classical sense of performance as a way of offering a situation, and being aware of how the situation is being received. That's what a lot of performance is, especially if it also borders some kind of theater. Sometimes there's a side of it that slightly bothers me. Especially some of the more academic new music concerts, where people fall asleep. I had several of those in a row. I got to think that maybe I was somebody's



Laetitia Sonami and the Lady's Glove, 1999. PHOTO BY ANDRE HOEKZEMA.

waitress, here to entertain somebody so they can eventually go to sleep! So I went into other things like sound installations.

Then I saw this movie by Almodóvar, *Talk to Her*, and there's a woman Toreador. It's funny, in my mind the scenes of her with the bull were a major element, but I went back to excerpt those scenes and I got at most five seconds. The tension between her and the bull really resonated with me—the gestures, the communication—and I felt really close to that idea. I thought again about performance, and what was it that made me watch her? You know, I'm a huge fan of John Bischoff's music, and he's working with a laptop, while on the other hand some people who use gesture controllers do not particularly interest me. So it's not really about creating some kind of embodiment of a performative action with electronics. It has to do really with the concentration, and the focus the performer radiates. That's when performance is beautiful. The intimacy that the performer presents to the public, and you witness this intimacy and you feel so fortunate. In a way the Lady's Glove allows a little of that concentration. So I went back with renewed excitement, and that is when I worked on the fifth

version. I don't know how long I'll do it, it has been a while now, but it has been interesting.

You mentioned the social commentary of it. Can you talk more about that?

Well, at first, there was this huge interest in virtual reality systems. On the entertainment side, people were using the Mattel Power Glove. It was very macho, which now I enjoy more, but at the time I was like, Come on . . . you know? [Laughs] This whole idea of macho control of technology, which shows itself so well in wars. So I thought, mine will be a sexy glove, sexy and feminine—'cause I'm French, and it's going to be a French glove! [Laughs] But with jokes like that, you think back, and if you're on that same "joke" for more than several months, you have to consider that it may not be a joke but something you're really serious about.

I always was amazed at what technology can do. I often feel that it is a projection of our dreams, illusions, desires, and it reflects how we see the world. I feel you have to respect that imagination. To reduce it to very crude systems is somewhat disrespectful of our imagination! I never thought of this while doing it, obviously; it was only later on, while reflecting on my obsessions. And I think there was always a desire to do something that would really display the magic, and refinement (or crudeness!) of the electronic world, and how delicate it can be. It just can change. It's like air, like wind. They are only moving electrons, which can completely change things around. So this desire to play with the subtlety of it—as opposed to reduce it—was maybe more feminine, you know? I've been interested by the fact that women can be more interested in subtleties of sounds than men are. If one wants to make really rough, crude generalizations! Men have done really interesting works that are system-based, very interesting systems, structurally fascinating, but sonically a little poor! [Laughs] I look at the work that was sonically incredible, and it's often been women, but of course not all.

What sounds have you typically worked with?

It's changed a lot, and has also gone through a lot of phases. When I first tried to play with sounds I was so puzzled, so I would listen a lot to the environment, try to figure out how sounds came together. I was

just blind in a way. So I would sit down and listen to sounds, and take notes: airplane comes, dog barks, etc., and just try to dissociate the various elements which made the landscape. A lot of the work I did was recording natural sounds and extracting from them. And then at some point, I got less interested with natural sounds; they became too familiar. So I became more interested in synthesis, and then it became more of a juxtaposition of abstract synthesis with sampled sounds, or the sampled sounds are processed so you can't really recognize them. I am interested in the juxtaposition of abstraction and iconic elements; there might be a voice or a breath, but it would happen at a time when the landscape is very abstract, so there is always this play back and forth between recognizable elements and abstraction. Lately I got a little tired of even using processed samples, because it's very hard to modify. There's only so much you can do, and you cannot take it apart that well. One new piece is almost all digital synthesis in Max/MSP, sending pulses through resonators and filters, filtering raw signals and composing new signals on the fly. It's allowed me to be a little more flexible, less predictable. We have so many possibilities of sounds.

It can be hard to choose.

It's hard to choose, but you're always attracted to a particular color or texture, and these attractions change depending on what you are going through, right? It's kind of like a bee with a flower. You don't know, but you just go and zoom to those particular sounds. And then you gather them over two, three months, and you listen to them, and they start telling stories.

When you perform, is it largely improvisational, or do you have compositional structures mapped out?

It's not so improvisational. For me it's really hard to improvise with the software I write, as opposed to George Lewis or other composers who make "listening" software that responds to events or musical gestures [see Lewis 2000]. In my solo performances, the sounds have been chosen, the mapping has been chosen, and the processes have been defined. In the piece I am working on now, I try to be much more fluid in the way things happen. But it's hard. Sometimes it becomes too random when I leave it more freedom. The way I've resolved the situation is with the use of templates. Within a template, I have a set

of possibilities: which sounds can happen, which sound is mapped to which sensor, are there filters, resonators, DSP of some kind . . . I have the freedom to move between templates and play with all those elements, which allows for a dynamic remapping of the sensors and the sounds. That seems to work pretty well, wandering around within templates. The only problem I've found, because I'm very much tied to a narrative form, as abstract as it may be, is the tendency I have to go in a very strict order between templates. And usually, because of the way I think of music, the templates are pretty defined, mental spaces. With the new piece, I'm trying to know less, but I'm struggling with this fact that I really want certain things to happen. It's not like they're recorded, but I have a very defined idea of what I want to happen. It's really very precise, kind of surgical.

Do you make recordings of your performances?

Well, I'm supposed to, but I have many problems with CDs. First, I have so little time, and I'm more interested in the situation of performance, which I find quite magical. You're alone, but then you're with people for a half hour, one hour, and you create this moment, and then that's it—it's gone, and nothing's left, just the memory of the moment. Sometimes it's good, sometimes it's bad . . .

Also, what attracted me to sounds is their impermanence and fleeting quality. The idea of fixing sounds, and thinking of sounds repeating themselves outside of a context, is so much against what I like about sounds, that I'm having trouble thinking about recording—the idea of “objective” sounds adaptable to any situation or time.

And the other thing is probably a lack of confidence. The idea of commitment of any kind terrifies me—that you would have to commit a sound to a particular time. It's like, Oh, God, how do I know it should be there? [Laughs] So there's that, and then there's just so much objects already in the world. But I'm glad other people are doing it. And one day I'll do it. I go through these arguments so many times. I was talking with this really nice composer, and he was saying, Just think of it as documentation, we want to hear it! And I said, But there's a hundred thousand other people's music that they could hear, or they can also come to my concert. And if you can't come to the concert, well, you just can't hear it! For some reason, musicians got into this idea that everything you do has to be recorded. If you use a

recording medium, fine. But if you're not, if you're doing performance, that is a very different situation.

:::

Sonami teaches at the San Francisco Art Institute and the Milton Avery Summer Program at Bard College and was the David Tudor Composer-in-Residence at Mills College in 2007.

Laetitia SONAMI
REQUIEM FOR the *LADY's GLOVE*
1991-2016

Around 2014 I decided to start retiring the lady's glove, the instrument I had used for the past 25 years. Here is looking back... and forth.

Over the years, the *lady's glove* merged with my left arm, an active layer between skin and muscle. Full of urgent needs, limitations, humor and failures, it became a language I could ride with my eyes closed ¹.

Sprung from satire ², the *lady* settled into an indispensable interface, inextricable from my imagination. What first started as dreams of control and virtuosic display, turned into dreams of chaos. From circus dog, it imagined itself roaming wide plains, yet still tethered by thick mesh of wires ³.

Instead of re-inventing herself, it stepped aside.

**- Virtuosity?... Are you doing anything?
No... just dreaming**

Sounds, events, emotions migrated to specific gestures, and some became fused:

The bending of the wrist (upwards / palm down) attached itself to mechanical sounds, functional or dysfunctional (pumping, breathing machines and various clicks);

Hands moving back and forth became changes in density or changes in acceleration as they scanned thru layers of sound deposits;

The bending and scribbling of all left fingers like a spider's legs, to speech utterances or small events near by ⁴;

The bending of just the index with arm extended, to recalling far away sounds such as pigs and screeching wheels;

Rotating the hand, palm to the sky, to halting all sounds or to an abrupt change in landscape (halting can also be associated with bending the elbow and moving forearm up: a traffic stop);

¹ Ride, as in horses.

² The first version of the lady's glove was built for a collaboration with Paul DeMarinis "Mechanization takes Command" (1991). Made from kitchen gloves, it was a response to the macho, robotic, grey rubber Power Glove from Mattel – a vision of women in the kitchen doing electronic music while washing dishes.

³ The twenty-eight or so sensors embedded in the black arm-length lycra glove (two ultra sounds, three accelerometers, seven bending strips, one pressure pad, nine micro switch, five Hall effects, one microphone) were first conditioned in Steim's Sensorlab and sent as MIDI input to Max-MSP where all the mapping would occur, hence the dark web of connecting patch cords. Michel Waiswiz (1949-2008), previous director of Steim and master of 'the Hands', seemed always puzzled at how inflexible my mapping strategies were...

⁴ I most often think of sounds as distances (see Arcana III -Musicians on Music by John Zorn: "Seizing a sound and smelling its belly to fit it in a folder").

Bending and moving the hand close to the ground, to what is always there: water, low frequencies or empty spaces;

Elbow back, palm up, hand by the waist while curling fingers, to something like “come on....” and on and on...

Gestures, sensors, sounds, sonic narratives and emotions engraved themselves in pre-arranged patterns, and became irresistible.

- How many buttons do you need?

0 buttons

- Not possible

Then give me 5, I will find where the edge is

A major issue in designing digital interfaces or hybrids is the apparent limitlessness of ‘materials’ – each thought has its own button. I do believe now that the instrument exists on the edge of its limitations – Those are not clear at first as the choices are vast, but eventually one needs to settle on a particular configuration, and play with its limits ⁵.

The hand does all kinds of things without its owner noticing. Small gestures that escape consciousness, small gestures that point to other worlds. These are not nervous ticks, they are the sounds that escape the main line, harmonics and air...birds.

- Where did you bury it?

Under the tree... with the dead dogs ⁶

I stopped playing the *lady's glove* because my imagination and the *lady's glove* had merged into one I knew what it could do and I knew what I could do

My hand became the camera, scanning thru sonic territories with its favorite angles; the angles needed a major shift

I did not want to stand onstage at ninety shaking my arms with trembling folds of flesh

Gestures, sounds and geographies slowly became mummified

I left it behind because I did not want to blemish it with poor looking associations ⁷

So...I leave gloves and other wearables to small dictators and corporate powers.

⁵ Not that I wish to advocate a paucity of resources, but the early digital music of the late seventies for example (with musicians like David Berhman, John Bischoff, Paul DeMarinis) and the 8 bits /1k memory of the KIM-1, seems to have a “palpable” materiality.

⁶ Not true

⁷ Recent gloves and their parametric monotony: do you really need to bend your finger to raise the pitch of your voice?

And now the **Spring Spyre**⁸— it started as a conversation, a wandering.

There were desires and some rules, dictated by the *lady's glove*⁹. After three years, it is starting to talk and it wants things. It is a bit more tyrannical, I do not expect we will merge. It sits like empty space and I stare at it, recalling the Oracle. I am trying to understand its language. I think the Spring Spyre wants a story, it wants characters¹⁰.

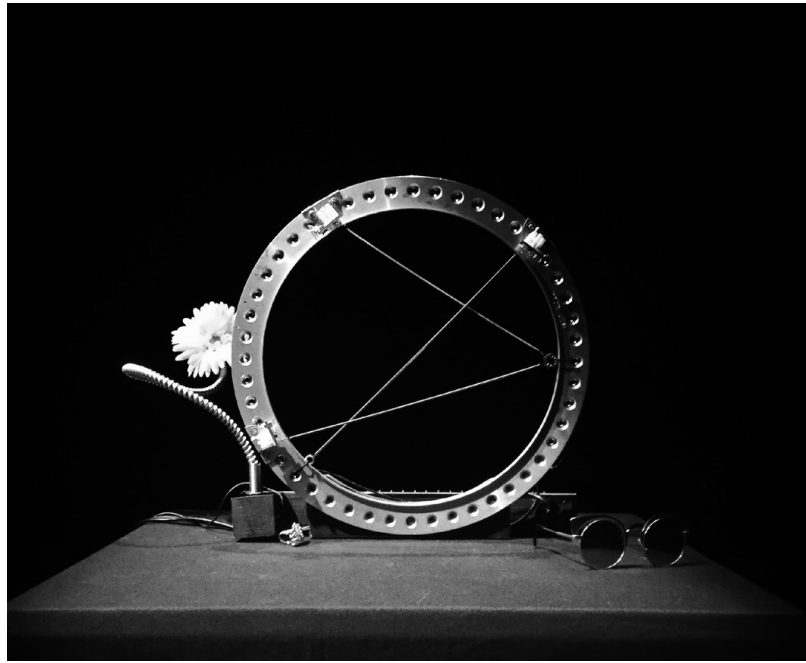


Photo by: Jean Piché

⁸ The Spring Spyre was named by Eliane Radigue who composed its first piece, OCCAM IX. It was built to allow for a more improvisational approach, and also more unpredictability, which is ironic as Eliane's music is all but unpredictable or improvisational. It currently has three audio pick ups stripped from reverb tanks. The three audio signals are analyzed (i.e. "feature extraction") and fed as inputs for training the neural nets in Wekinator (Machine Learning software by Rebecca Fiebrink). The nets control the audio synthesis in real time in Max-MSP. A bank of faders (from a hacked Peavey PC 1600) allow for the mix of the various synthesis. The range of unpredictability can be adjusted by the "width" of the training.

⁹ The new interface needed to be easily replaceable, pass TSA screening smoothly, generate a more complex sensing signal, and again, be somewhat unreliable.

¹⁰ I am currently developing a story about three women living in Peripheral Vision, a small town off of route 95, inspired by suburban women and night scientists. The women eventually merge after an earthquake strikes the town.



Photos by: Frank Baldé

